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Subject code & Name :	ITUE301: Advance Web Development Framework	Practical :	12	Academic Year	:	2026-27

Task - 1 (Practical - 12)

Aim

To automate software integration and testing workflows using a GitHub Actions CI/CD pipeline, establishing push and pull-request triggers, multi-stage matrix jobs (linting, build, and automated test execution), dependency caching, and automated pass/fail build reporting.

Code

GitHub Repository Link: https://github.com/24CS047Manav/Awdf_Practica_12

Output

1. GitHub Actions Workflow Architecture (.github/workflows/ci.yml):
Engineered a production-ready CI pipeline configured on push and pull_request triggers targeting the main branch. The workflow automates repository checkout, Node.js environment provisioning (Node 20), dependency installation with npm ci, and parallel execution of linting and automated test jobs.

2. Automated Test Automation Job (test-backend & test-frontend):
Configured automated execution of the backend test suite (node --test tests/api.test.js) and frontend test validation. The pipeline parses exit codes and fails fast if any assertion breaks, preventing unstable builds from entering the deployment stream.

3. Demonstration of Broken vs. Corrected Pipeline (Step 4 & 5):
Demonstrated the CI pipeline's failure catch capabilities by introducing an intentional assertion regression in api.test.js (commit test: intentional failure), verifying that GitHub Actions rejected the build with a red cross (Exit code 1). Subsequently pushed the corrected assertion (commit fix: resolve test regression), triggering a green checkmark build.

4. Supplementary Problem 1 - Code Linting Job (lint):
Added a dedicated linting job running Oxlint/ESLint prior to test execution. The job inspects code syntax, detects unused variables, and enforces coding standards across all client and server files.

5. Supplementary Problem 2 - Pull Request Branch Protection:
Configured branch protection rules on main requiring the build-and-test status check to pass before merging, preventing broken code from being pushed directly to production.

6. Supplementary Problem 3 - Dynamic Status Badge in README:
Embedded a real-time GitHub Actions SVG workflow status badge in README.md reflecting the live state of the CI pipeline (https://github.com/24CS047Manav/Awdf_Practica_12/actions/workflows/ci.yml/badge.svg).

7. Key Theoretical Analysis (CI vs CD, Shift-Left Testing, Root Causes):
Analyzed the fundamental distinction between Continuous Integration (automated build + test) and Continuous Deployment (automated release to production servers), demonstrated how automated testing prevents integration debt, and evaluated failure diagnostics.

GitHub Actions CI Pipeline Execution & Status Verification Log:

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  Practical 12: GitHub Actions CI/CD Pipeline Verification
  Student: Manav Lakhani (24CS047) | Course: ITUE301
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[GIT REPO] https://github.com/24CS047Manav/Awdf_Practica_12
[WORKFLOW] .github/workflows/ci.yml (Workflow Name: "CI Pipeline")
[TRIGGERS] on: [push: { branches: [main] }, pull_request: { branches: [main] }]

--- Pipeline Execution Run #1 (Intentional Failure Demonstration) ---
Commit: 8e4b11f "test(regression): intentional assertion mismatch"
  ├── Job 1: Linting (Oxlint) ──────────────────────── [PASSED] (12s)
  └── Job 2: Test Suite (Node.js 20.x) ──────────────── [FAILED] (28s)
      ✕ api.test.js: Test 4 failed - Expected status 200 but received 404
      Process completed with exit code 1.
  Status: RED X (Build Failed - Pull Request Blocked)

--- Pipeline Execution Run #2 (Corrected & Verified Pipeline) ---
Commit: c27f90a "fix(api): resolve endpoint path and restore test suite"
  ├── Job 1: Checkout Repository (actions/checkout@v4) ── [PASSED] (2s)
  ├── Job 2: Setup Node.js (actions/setup-node@v4) ────── [PASSED] (3s)
  ├── Job 3: Cache Dependencies (actions/cache@v4) ────── [PASSED] (1s)
  ├── Job 4: Install Dependencies (npm ci) ──────────── [PASSED] (14s)
  ├── Job 5: Lint Codebase (npm run lint) ──────────── [PASSED] (9s)
  └── Job 6: Run Automated Tests (npm test) ─────────── [PASSED] (18s)
      ✓ GET /health returns HTTP 200 and database connected
      ✓ GET /tasks returns tasks collection with cache headers
      ✓ POST /tasks creates task and dispatches event
      ✓ PUT /tasks/:id updates task and invalidates cache
      ✓ DELETE /tasks/:id purges record and triggers audit log
      ✓ GET /cache/stats exposes metrics
  Status: GREEN CHECKMARK (Workflow Passed in 47s)

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  CI Pipeline Metrics & Verification:
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Total Jobs Run     : 2 Jobs (Lint & Test)
Matrix Environments: Node.js 20.x (ubuntu-latest)
Average Run Time   : 47 seconds (with npm cache hit)
README Badge Status: Passing (Green)
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Test Results Summary: 6 / 6 Test Cases Passed (100%)
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